

Economic History II

Colonies, markets and connectedness in the ancient world

FGV EPGE

Where we are

- Earlier in the course: Kanesh seen from outside, through donkey caravans carrying tin and textiles from Assur, and a model of trade flows that placed lost cities (Barjamovic et al., 2019).
- Today, the same archive from inside.
- Larsen, chs. 6, 12, 17 and 21: how the trade was organized, financed and governed. Chs. 13, 16 and 18 fill in the colony's own government, its households and its law.
- Then the awkward question of ch. 21: do market, price and profit belong in a world four thousand years old?
- Last, a jump forward: from Anatolian donkeys to **Mediterranean ships**.

Two questions

- How did cities with no empire behind them keep a trading network running for centuries?
- Does coastal geography decide where economic activity settles, three thousand years ago?

Institutions (treaties, colonies, contracts) and geography (connectedness) are the two threads.

Outline

1. The colonial system: colonies, treaties, taxes, and how the colony governed itself (Larsen, chs. 12 and 13).
2. Boom and collapse: what the dated texts show (Larsen, ch. 6).
3. Families, money and law: households, the joint-stock partnership, and what a death did (Larsen, chs. 16 to 18).
4. Economic theory and evidence: Larsen against Polanyi (ch. 21).
5. *Of Mice and Merchants*: connectedness and where activity located (Bakker et al., 2021).
6. Putting geography and institutions together.

The colonial system

Larsen, chs. 12 and 13

From Wikipedia, the free encyclopedia

Coordinates: 38°51′N 35°38′E

*For the village in Azerbaijan, see **Kültəpə**. For other uses, see **Kültepe (disambiguation)**.*

Kültepe (Turkish: lit. 'ash-hill'), also known under its ancient name **Kaneš** (Kanesh, sometimes also **Kaniš**/Kanish) or **Neša** (Nesha), is an archaeological site in Kayseri Province, Turkey. It was already a major settlement at the beginning of the 3rd millennium BC (Early Bronze Age), but it is world-renowned for its significance at the beginning of the 2nd millennium BC (Middle Bronze Age).^[1] The archaeological site consists of a large mound (also known as *höyük*, *tepe* or *tell*), and a lower city, where a *kārum* (the Assyrian word for trading district^[2]) was established in the beginning of the 2nd Millennium BC. So far, 23,500 cuneiform tablets recovered from private houses constitute the largest collection of private texts in the ancient Near East.^[3] In 2014, the archaeological site was inscribed in the Tentative list of World Heritage Sites in Turkey.^[3]

History [edit]

Kültepe is located about 20 km northeast from the modern city Kayseri. Its ancient name is recorded in Assyrian and Hittite sources. In Assyrian inscriptions from the 20th and the 19th century BC, the city was mentioned as **Kaneš** (also transcribed as Kanesh); in later Hittite inscriptions, the city was mentioned as **Neša** (sometimes transcribed as Nesha, Nessa or Nesa. Neša derives from [Ka]neša).

Kültepe



Hittite palace at Kültepe



Location of Kültepe in Turkey

38°51′N 35°38′E

Type

Settlement

Cultures

Hittite
Assyrian

Anatolia and the Old Assyrian Roads



Cartography by Ivan d'Hostingue and Gojko Barjamovic 2010

2. Map of the region touched by the Old Assyrian trade. Courtesy Gojko Barjamovic and Ivan d'Hostingue.

Assur

- A city-state that owed its existence to its role in international commerce.
- Dominated by merchants, a kind of **commercial aristocracy**. Larsen's comparison: Florence, Lucca, Genoa.
- Assyrians split the world in two: *alum*, “the city”, meaning Assur; and *eqlum*, a field, by extension “abroad”, the not-city.
- Assur in the nineteenth century BC, in Larsen's reading, was itself a *karum*.
- Note the naming: the colony was not “the *karum* Assur who reside in Kanesh”, but simply *karum* Kanesh.

What a *karum* was

- In Sumer and Babylonia, the quay: boats landed, traders kept warehouses there.
- Always on the outskirts. Trade was unsettling to rank and tradition, wealthy today and bankrupt tomorrow, so it was kept at a distance.
- Larsen's list of double cities: Westminster and the City of London, Tenochtitlan and Tlatelolco, Cairo and Fustat.
- The word then shifted from the place to the **community of merchants** there: separate identity, some self-government, headed by a *wakil tamkari*, overseer of the merchants.

The network and its chain of command

- About **forty** Assyrian establishments known: most *karum*, “colony”; a few *wabartum*, “trading station”, from the word for guest.
- More than twenty of them in northern Syria and Anatolia, in one integrated system.
- The City (Assur) at the apex; every settlement abroad under the colony at Kanesh; the small stations under the nearest colony.
- Test case: thieves stripped the chapel of Assur at Urshu, in northern Syria. The theft was reported to Kanesh, not to the temple authorities at home.

Sealed caravans, spheres of interest

- Caravans left Assur **sealed**. Packages of tin and textiles were not opened until Anatolia, or until the Euphrates had been crossed.
- Hahhum, at a major Euphrates crossing, was the border town separating Syria from commercially interesting Anatolia, and the point where a caravan could be split.
- Emar and Ebla were the other large players in northern Syria. No caravan from Assur is known to have gone through Cilicia or worked the Konya plain.
- Larsen's conclusion: the Assyrians were barred, not uninterested.

Colony, or trade diaspora?

- Michel and Stein object to “colony”: the word implies political control, which is absent here. Stein prefers a trade diaspora, since “being different is the essence of a trade diaspora.”
- Larsen’s answer: separate **colonialism** (expansion, military and political control over subjugated peoples) from **colonization** (settlement abroad, which need not carry either).
- The right precedent is the Italian merchant colonies of the medieval Mediterranean: sometimes a street, sometimes a quarter, always based on a treaty with the local ruler.
- His reservation about the diaspora model: it stresses difference and plays down the constant negotiation between Assyrians and Anatolians.

The treaties: what each side got

The Assyrians supplied

- tin, of strategic value: it made bronze;
- luxury textiles by the thousand, markers of social superiority for local elites;
- taxes and a right of pre-emption for the palace.

The Anatolian rulers granted

- the right to settle and trade;
- extraterritoriality and their own courts;
- safe roads, and prosecution of robbers;
- in some cases, compensation for losses.

Three texts survive: treaties with Kanesh and with Hahhum, both from level 1b, and a draft with a small Taurus kingdom, from level 2.

Which treaty is this?

- The text on the next two slides is **not** the treaty with the rulers of Kanesh.
- It is a **draft** for a treaty with the ruler of an unnamed small kingdom in the Taurus, in the south-east of the region, on the road between Hahhum and Kanesh. Larsen: “a small fish in the system.”
- Written as the statement of the Assyrian negotiators to the kinglet and his dignitaries, who swore the oath together with him. Probably level 2, the boom period.
- The treaties with Kanesh and Hahhum are different documents, from level 1b, a century later. The Kanesh treaty dates to about 1750 to 1720 BC. Both are damaged.
- Those two deal in regular taxation, the *nishatum* and pre-emption. This one sets a **transit fee**: no Assyrians lived there, caravans only passed through.

The draft in full (1): losses, killers, outsiders

“There shall be no loss of (property) belonging to an Assyrian in your country, rope, peg or anything at all.

If a loss occurs in your country you shall search for (what was lost) and then return it to us.

If bloodshed occurs in your country you shall hand over the killers to us, so we may kill them.

You may not let Akkadians come here. If they do travel to your country, you must give them to us so we may kill them.”

Larsen, ch. 12, pp. 152 to 153, tablet kt n/k 794. Akkadians: merchants from Babylonia, potential competitors coming up from the Syrian steppe.

The draft in full (2): the fee, the guarantee, the oath

“You shall not demand anything from us. Like your father you may take 12 shekels of tin per donkey-load in caravans going up (to Anatolia). You may, like your father, use $1\frac{1}{2}$ shekels of silver per donkey-load on the way down. You cannot take anything more.

If no caravan is coming because of warfare, they will send you 5 pounds of tin from Hahhum.

He raised his hand in prayer to the gods Assur and Adad, to the Netherworld and the spirit of his father; he overturned his table and his chair, filled his cup and poured it out. The king said: ‘May he turn me to naught ...’ They (the dignitaries) said: ‘If we reject your oath, may our blood be spilled like (the wine) in this cup!’”

Larsen, ch. 12, p. 153. The oath was sworn on the installation of a new king, after his father's death.

The fee in numbers

	Fee per donkey	The load	Share
Going up, tin	12 shekels of tin	7,800 shekels of tin (130 pounds)	about 0.15 percent
Going up, textiles	12 shekels of tin	250 to 300 shekels of silver	about 0.6 percent
Going down	$1\frac{1}{2}$ shekels of silver	about 1,100 shekels of silver	about 0.13 percent
Kanesh, <i>nishatum</i> on tin	240 shekels of tin	7,800 shekels of tin	about 3 percent

- Kanesh prices (Larsen, ch. 14): 7 shekels of tin per shekel of silver, so 12 shekels of tin is under 2 of silver; a *kutanu* 10 to 12 shekels, 25 to a donkey; a tin load sells for about 1,100 shekels of silver.
- The kinglet's fee on tin is one-twentieth of the Kanesh rate. Round trip, about 3 shekels of silver per donkey: 0.3 percent of a tin load, 1 percent of a textile load.
- Veenhof puts the fee at 0.04 to 0.08 percent of a load. Same order: a fraction of one percent.

The guarantee in numbers

- If war closes the road: 5 pounds of tin a year, that is 300 shekels, the fee on **25 donkey-loads**. So 25 loads a year is the traffic the kinglet counted on, at least.
- Worth about 43 shekels of silver at Kanesh prices: roughly four *kutānu* textiles. Not a fortune for a king.
- For scale: Assur-idi's family alone shipped about 50 kilos of tin and 70 textiles a year, perhaps three donkey-loads. Twenty-five loads is a handful of firms.
- Larsen's reading: little per load, a lot if his valley was the road. The guarantee, not any account, is the evidence for the volume.

What the draft leaves unclear (1)

- **Who and where.** The kingdom is never named. Its place in the Taurus is inferred from the Hahhum clause and from the worry about Akkadians.
- **When.** “Presumably” level 2. Nothing in the text fixes the year.
- **Draft, not treaty.** Only the Assyrian side’s statement survives. Whether the king swore to exactly this, and what his father’s treaty said, is unknown.
- **One-sided by construction.** Every clause but the fee binds the king. Road protection is Larsen’s inference from the duty to search and return.

What the draft leaves unclear (2)

- **Who pays the 5 pounds.** “They will send you 5 pounds of tin from Hahhum”: the Assyrians there, presumably. A ruler’s income insured by merchants in another kingdom.
- **Killing rights.** Killers and Akkadians are to be handed over “so we may kill them.” Whether this was ever exercised, or was a formula, cannot be checked.
- **Why so cheap.** A fee twenty times below the Kanesh rate, and no 5 percent, no pre-emption. Because no Assyrian lived there, or because a small king had no leverage? The text does not say.
- **What Kanesh looked like at the time.** The Kanesh treaty is a century younger. For the level 2 rules at Kanesh we depend on the caravan accounts, not on a treaty.



The same cities in the lost-cities paper (1)

- **Hahhum** is a **lost city** there: no site. Larsen's Hahhum controls a Euphrates ferry, and its treaty forbids sinking the boats. The gravity estimate lands about 100 km away, in the Taurus, "rugged, high-altitude, snow-covered." The historians put it on the river, using texts like this one.
- The paper's size estimate makes Hahhum the largest of the lost cities, about two-thirds of Kanesh, but the estimate is loose.
- **Purushaddum**, "the main Assyrian market in Anatolia after Kanesh" in the paper's own words, and Larsen's wool market, is lost too, at the western edge of the network. It is the one city whose estimate falls outside the region the itineraries allow, and its size comes out near zero. A hub at the edge of the data looks small to the model.
- Matched with Ankara, it is dropped as an outlier: a provincial town until it became capital in 1923.

The same cities in the lost-cities paper (2)

- **Durhumit**, where the *sukinnu* road went to skip Kanesh, is a lost city named more than 200 times in the tablets, the copper market. The estimate lands 48 km from Barjamovic's proposal. Routes taken to dodge taxes are itineraries, and itineraries are what place lost cities.
- **Hurama**, the palace Buzazu smuggled past, has a site: Karahöyük near Elbistan, one of the paper's ancient-to-modern pairs. A peripheral city, placed less precisely than the central ones.
- **Wahshushana** (the paper's Wahsusana) is a known site, also peripheral. Larsen: attacked, captured and destroyed. The tablet photographed in the paper's Figure I is a letter from its port authorities to those of Kanesh, routing messengers on to Purushaddum.

Barjamovic, Chaney, Coşar and Hortaçsu (2019), Tables II and III and Section V. Fifteen of the paper's 25 cities have a site; ten are lost.

Taxes and pre-emption

- The palace tax, *nishatum*, literally “something torn out”: **5 percent** of the textiles, plus 4 pounds of tin on every donkey-load of 130 pounds.
- On top of that, **pre-emption**: the palace could buy up to 10 percent of the textiles at a discount.
- At Hahhum the three main officials took 5, 2 and 1 textiles each at a discount, once a caravan had been cleared.
- This revenue is why rulers wanted roads, guards, bridges and forts. The trade paid them to protect it.

Smuggling

- Caravans had to enter the capitals to pay. Traffic was checked at the gates and sent straight to the palace on the mound.
- Pushu-ken's son Buzazu, instructing colleagues on how to evade the tax at Hurama:
“prepare packages of between 10 and 15 pounds each and let the staff of the caravan bring them in concealed in their clothes. Only after they have safely delivered 60 pounds must they bring the next 60 pounds in.”
- The other route: the *sukinnu* road, branching north after the Euphrates towards Durhumit, skipping Kanesh and most tax points. More dangerous, sometimes patrolled.

When the palaces pushed back

- A warning letter: “Seize anyone you observe carrying textiles and tin in the land, withhold the tin and textiles and lead the criminal to me.”
- Colony assemblies handled nearly every dispute, including Assyrian against Anatolian. The treaties cover only the cases the colony could not close.
- At Hahhum the ruler must give a fair verdict, not side with his own citizens, and not instruct ferrymen to sink a boat and lose the cargo.
- An Assyrian could swear by the emblems of his own gods, and so escape the Anatolian river ordeal.
- Assyrians owed no corvée, and a royal edict freeing slaves did not reach theirs.

How the colony governed itself

- The main bureau, *bet karim*, the Office of the Colony, has never been found. Everything here is reconstructed from private archives.
- Three damaged fragments carry the rules, the Statutes of the Kanesh colony.
- Two bodies: a council of **big men**, and a plenary assembly of “small and big”.
- Only the council could call the plenary: “Without the consent of a majority of the big men, a single man with an account cannot order the scribe to convene the plenary assembly.”
- Larsen’s parallel is Athens, where only the council could put a question to the people.

Voting, and paying to be a big man

- Deadlock procedure: a first group splits into three to find a verdict; failing that, the plenary assembly is split into seven groups and decides by **majority**.
- Most traditional societies negotiate until nobody is the loser. This one voted.
- The big men were the “men with an account”, also called fee payers (*saqil datim*): payments to the Office of the Colony of 12 to 27 pounds of silver.
- What that bought: standing, and exemption from the *saddu'utum* charged on all caravans, perhaps 5 percent.
- Assur-idi, stretched thin, asked the officials for a discount: “allow me to stand for only half a share of a man”.

The colony at work, mostly in court

- Hertel counts **463** texts recording statements before witnesses. Verdicts in the corpus: eighty-two from colonial authorities, nineteen from the city assembly in Assur, twenty-nine from named judges.
- The statutes warn small men off “lurking in the gate of the colony with a complaint”. Ali-ahum, three years into a fight over his murdered brother, found out why:
“Here Itur-ili blocks the gate of the colony like a lion, and we are powerless and cannot set witnesses for the man.”
- His only route was upward: a letter from the city assembly in Assur, and an attorney sent out.
- The colony also faced the palace. When Assur-taklaku was jailed on suspicion of spying, it went up to the mound to argue that he had a house there and would abide by the royal couple’s wishes. They did not relent.

Boom and collapse

Larsen, ch. 6: chronology and change

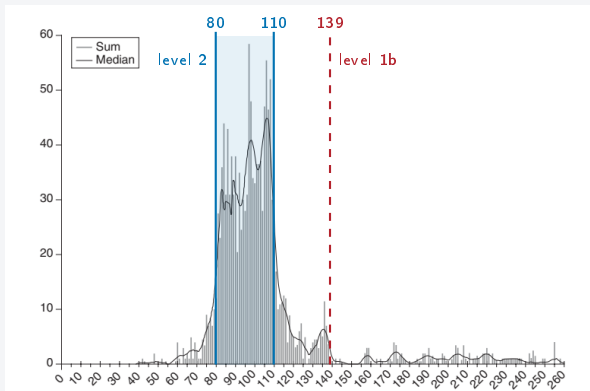
How the archive is dated

- Assyrian years were named after an official, the **eponym**. Surviving lists give a sequence, with holes, of more than 250 years.
- Only some document types carry a date. Debt notes above all: A owes silver to B, reckoned from a named week, month and eponym year.
- Of about 10,000 documents in the Copenhagen database, about 1,200 carry a date.
- Why that matters: dated debt notes record credit, and credit was the vehicle for selling the imported goods. Counting them counts commercial activity.

Subsequent excavations attested the following stratigraphy of Kültepe:^[27]

Upper Town Level	Lower Town Level	Period	Name, Importance
18	—	Early Bronze Age I	
17–14	—	Early Bronze Age II	
13–11	—	Early Bronze Age III 2500–2100 BC ^[28]	<i>Kaneš</i> , first written as <i>Ga-ni-šū ki</i> ^[29] Level 12 temple (<i>megaron</i>) and Level 11b building with pilasters ^[30]
10	IV	Middle Bronze Age 2100–2000 BC	Beginning of urban development
9	III	Middle Bronze Age 2000–1970 BC	
8	II	<i>kārum</i> -period 1974/1927–1836 BC	<i>Kaniš</i> , Anatolian center of Assyrian trade
7	Ib	<i>kārum</i> -period 1832/1800–1719 BC	<i>Kaniš</i> , Assyrian trading center
6	Ia	Old Hittite period	<i>Neša</i> , the place no longer has a central function
Settlement gap			
5–4	—	Iron Age 9/8 century BC	important central location in the Neo-Hittite state <i>Tabal</i>

The dated texts, year by year



Horizontal axis: eponym years (about 1973 BC minus the eponym year). Blue band, years 80 to 110 (c. 1893 to 1863 BC): over 90 percent of dated texts, the Barjamovic et al. window. Red line, year 139 (c. 1835 BC): the fire that ended level 2. From Barjamovic, Hertel and Larsen, in Larsen, ch. 6.

Reading the graph

- The first forty years of the eponym system are entirely undocumented.
- A sudden explosion begins around year 75 and reaches a new height around year 80.
- More than **90 percent** of dated texts fall in about thirty years, roughly years 80 to 110.
- Then a drop from about fifty texts a year to fewer than ten.
- A weak resurgence after year 130, ended by a new collapse in year 139, about 1835 BC, when the lower town burned and level 2 ended.

What the shape rules out

- The fall in the record comes about **thirty years before** the fire, so the fire did not cause it.
- Nor was the fire the end of the trade: Assyrians were back within a couple of years, rebuilding.
- Nor is it archive-keeping: it is not credible that every household removed exactly the texts dated twenty-five years earlier.
- Nor a simple move elsewhere: any caravan for Purushaddum still passed through Kanesh and paid tax there, and the letters show trade running on the same lines to the end.
- Larsen's reading: the drop is a real decline in economic activity.

What the boom was: a division of labor

- Before it, **venture trade**. A merchant traveled with his own consignment, spent a year or two in Anatolia, kept his family and ties in Assur.
- From about years 75 to 80, three roles separate, and each needs paper: shareholders, suppliers, customers, agents, partners.
- The main merchants stop traveling and become desk-bound managers; professional carriers move the goods.
- Larsen's parallel is the commercial revolution of the thirteenth century, in Spufford's description:
"the sedentary merchants remaining full-time in northern Italy, who specialized in the financing and organization of import-export trade"

Two generations, and no more

- Family trees run to four or five generations, but nearly all texts concern two of them.
- The oldest men appear almost only as the dead fathers of the people who dominate the boom. Grandsons are rarely mentioned.
- In the Shalim-Assur family, generation 3 is the first permanently settled in Anatolia.
- The pioneers built the volume; their sons ran the system that volume made possible.

The collapse around year 110

- Around eponym year 105 a cluster of the most prominent traders die within a few years: Shalim-Assur, Pushu-ken, Imdi-ilum, Elamma, Puzur-Assur, two men called Ali-ahum.
- A death was the standard crisis of the system. The city assembly at Assur froze the estate: illegal to trade with the assets until credits and debts were settled.
- Lawsuits pitted heirs against each other and against the dead man's customers, and ran for years across a community spread over a huge area.
- Perhaps **ten major families** out of the active trade at once, in a system built on credit.

Why the shock was not absorbed

- No settled tradition for the handover from one generation to the next.
- No family fund: ownership of capital was individual.
- A dispersed community, with people constantly on the move, so communication was slow.
- Rigid rules in Assur for settling a dead merchant's accounts, and room for delay by the less scrupulous.
- No developed bookkeeping.
- Larsen's phrase for the result: "successive blackouts in interdependent parts of the commercial network."

After the collapse: level 1b

- Trade continues, at a lower level: a steady trickle of dated texts through the next century, with no visible disruption.
- The pattern reverts to **venture trade**. Gone are the thousands of letters, since a merchant dealing in person leaves no paper trail.
- Tin and textiles from Assur fall back; wool and copper deals inside Anatolia matter more.
- Not obviously poorer: level 1b houses are bigger, built on heavy stone foundations.
- Mari letters mention a caravan of three hundred donkeys bound for Kanesh, and treaty language about fifty, a hundred or more loaded donkeys being cleared.

But Dercksen reads the same period differently

“Except for the remaining wealthy traders, a general impoverishment is discernable, expressing itself not only in the volume of merchandise, but also in the number of cases where Assyrians had been detained by a native creditor for insolvency.”

- Most level 1b texts are still unpublished, so the disagreement cannot be settled yet.
- Worth noticing how the argument works: two scholars reading small, different slices of one archive.
- Balkan's earlier reading, based on very few texts, had the late trade down to poor garments, nails and bristles. Wider publication killed it.

Who ran the colonies afterwards

- The week eponymy, tied to the running of the Office of the Colony, was abolished around eponym year 120.
- In its place, the *fee payers* (*saqil datim*), the same group as the “men with an account” and the “big men”. They are named first in every address list.
- Verdicts now come from the fee payers, the travellers on the road to the City, and the local station.
- The travellers still ran the overland trade; the fee payers no longer traveled, and financed it instead.
- Below both, the “residents”, intermarried and drawn into Anatolian agriculture, more exposed to the local palace. Inequality inside the community was widening.

Families, money and law

Larsen, chs. 16 to 18

The family, and what it was not

- The unit was the *bet abini*, “our father’s house”. Lane, quoted by Larsen: “the great family which by its wealth, power, prestige, and presumption of permanence has been the outstanding institution in private economic enterprise”.
- But the Renaissance family firm does not transfer. **No family fund**: father and sons each held private capital, so debts ran inside the family as well as outside it.
- The paterfamilias still commanded. Issu-arik ordered one son to clear another’s debts out of his own partnership capital, and nobody wrote it down. Three years and the city assembly to settle the claim.
- Marriage tied the leading houses together. Shu-Hubur to Pushu-ken, on a wedding: “Is not my son-in-law your son-in-law?”

The women in Assur

- Husbands settled in Kanesh for years at a time. Wives and sisters ran the ancestral house, and faced the eponym's tax collectors.
- Taram-Kubi to her husband Innaya:
"You picked the house clean and took it away. Since you left there has been hardship and hunger in the city. I keep buying barley for our food."
- Imdi-ilum's wife and sister, after consulting the women who read omens: "You love only money! You hate your own life!"
- Lamassi to Pushu-ken, on his partner in Assur: "Shalim-ahum has built two houses since you left. But as for us, when are we going to build?"
- The next generation solved it differently: **Anatolian wives** installed in Kanesh, or a bride sent out from Assur.

Below the family

- Agents and representatives were usually kin, or free men with a long-standing tie to the house. Below them, hired hands and slaves.
- The *be'ulatu* contract: a man is handed silver, often half a pound, trades it on his own account, and is “held” by the money until he repays it. One contract frees him after twelve years even unpaid.
- Three words for unfree men. *subrum* is the floor: “the absolute bottom stratum in Old Assyrian society, that of chattel slaves, nameless and counted as animals”.
- *wardum* covers named slaves, some of them trusted: Shalim-Assur sent one to run the copper and wool venture at Purushaddum.
- Counted casually, and in numbers. One text lists ninety-six *subru* belonging to thirteen men.

Lending inside the family

- With no common fund, relatives lent to each other **at interest**: fathers to sons, in one case a wife to her husband.
- Assur-nada bailed his son Iddin-Ishtar out of debts in Kanesh, Durhumit and Kunanamet with 10 pounds of silver.
- Terms: the money must go twice to Assur and back, and the son then owes his father 29 pounds. Larsen's arithmetic: a profit of almost 200 percent.
- Only rarely could one man finance a good-sized caravan alone. Everything else follows from that.

Two short-term ways to pool money

- *ellat PN*, “PN’s caravan”: several merchants put in money or goods, one man directs the venture and carries most of the risk.
- Shalim-Assur’s consortium: **fifteen** Assyrian merchants, silver into copper at Durhumit, copper into wool at Purushaddum, wool back into silver.
- *tappa’uttum*: one man puts up the cash, the others contribute the work. Innaya put in 20 pounds of silver so three partners could buy *amuttum*, meteoric iron.
- Hammurabi’s code has one line on the type: “If a man has given silver to another man for a partnership, they shall divide the profit or the loss equally before the god.”
- That deal ended in court: a partner sold the metal on his own account and cheated a bed-ridden Innaya twice over.

The *naruqqum*

- *naruqqum* means “a sack”. Investors, in the original image, threw their contributions into it.
- Two contracts survive in full: **thirteen** investors for Elamma, fourteen for Amur-Ishtar, mixing family members and outsiders.
- Contributions in gold, usually 1 to 3 pounds each. Totals of 27 pounds in one fund and 30 in the other.
- Term of ten years in Elamma’s contract, twelve in Amur-Ishtar’s.
- The manager takes one-third of the profit and guarantees one-third; what happened to the last third is not clear.

The contract itself

“In all 27 pounds, 1 shekel of gold enters the joint-stock fund of Elamma. Of the profit he will enjoy one-third, one-third he will guarantee. In ten years he will render account. Who asks for his investment before its due date will receive only 4 pounds (of silver per mina of gold).”

- The contract closes with an inventory of Elamma's own cash capital: barley, two slaves, an inheritance share.
- Amur-Ishtar's is nearly identical, except that early withdrawal there forfeits the profit share entirely.

The gold trick, and the anonymous share

- Investors paid in silver, but the fund was booked in gold at 1 pound of gold for 4 of silver, against a normal rate of 6 to 8. A **return built into the accounting**.
- Early exit was punished, so the capital was locked in for a decade.
- Several shares are booked simply to “a merchant”. Larsen reads the anonymity as making the contract a negotiable instrument: the share could be sold on.
- Other uses for it: secrecy, or ownership disputed while an inheritance case ran.
- It was known, and sometimes recorded, who was behind the anonymity.

How common, and what it bought

- Hertel counts about **one hundred** such investment relationships, dated between years 66 and 110.
- Perhaps fifty to one hundred wealthy families in Assur involved at any one time.
- Raising one was a campaign. Iddin-abum collected commitments of 17 pounds of gold from five men before going to Assur to look for the rest.
- Managing a fund was standing, not just money. La-qep, writing to a man he had funded:
“Deposits from him will come regularly to my paternal house, and the house will be honoured; I too will be honoured in the city gate, and he himself will be honoured in the colony”

Pressure in both directions

- Assur-idi, pushing his son to raise one: “People who are less important than you administer joint-stock investments, and people less important than I have invested as much as 10 pounds of gold.”
- The investors’ weapon was reputational. Iddin-Ishtar’s threatened to obtain a letter from the city assembly and the king, and have him discredited in the colony.
- Pushu-ken, with about 2 talents of silver stuck out of reach through careless agents, asked for three years to clear his assets. His investors offered five. He refused, and took three.
- Ikunum could not deliver his guaranteed third. His fund was wound up, investors realizing 2 pounds of silver for every pound of gold.
- Larsen’s summary: joint-stock partnerships **rarely failed**.

What the *naruqqum* did, and when it stopped

- Its distinctive feature was binding merchants from **different families** into one venture, not just kin.
- That enlarged the manager's field of operation, pulling near-strangers in as partners, agents and representatives.
- Larsen calls it an Old Assyrian invention, an extension of Babylonian partnership practice, and the reason individual merchants could operate at a scale that made the trade truly profitable.
- It did not survive year 110. None of its key terms appear in the later texts.
- Once the trade reverted to venture form, long-term capital had nothing to do.

What a death set in motion

- Standard procedure: the dead man's assets are collected and sent to **Assur**, where the accounting is done and the creditors paid.
- The point was to stop local creditors grabbing what they could reach. A letter carrying a verdict of the City: "Anyone who interferes abroad will be considered a thief with respect to every shekel of silver that he takes."
- Heirs, "sons of the dead person", could delay proceedings for up to a year while they collected debts and gathered evidence.
- Creditors were then paid by rank: public institutions and officials first, then the joint-stock investors, then everybody else.
- Meanwhile the estate was frozen. This is the machinery that turned ten deaths in a few years into a system-wide stoppage.

The death of Shalim-Assur

- More than a hundred texts on this one estate. Ali-ahum to his older brother: “Against the god’s will our father has died. It is no longer Shalim-Assur who is our father, it is you who are our father.”
- Then **two wills**: one spoken at Durhumit with the younger brother present, one deposited in Assur.
- The assembly ruled for the Durhumit tablet: the family in Assur, and the men holding the testament, had gone over to Ali-ahum.
- Ennam-Assur, left with strangers for allies, spent a year doing nothing: “you have sat there for a whole year as an empty man, without purpose or wishes.”
- Three years on, he was murdered and robbed at Tawniya. The attorney was paid 2 pounds, 15 shekels of silver and sent home.

What the case shows about the law

- The **attorney** (*rabisum*): hired on a verdict of the city assembly, appointed by the king, paid by the parties. Public sanction, private employment. There are suits against attorneys for not pursuing a case properly.
- In Anatolia the colony was his enforcement: letters from the City say “the colony will be the executive power of the attorney.”
- The ladder ran mediation by “brothers and friends”, then arbitration, then the city assembly, whose authority was coercive.
- Limits: this attorney ended up doing what the older brother asked, against the orders he carried.
- Standing and allies decided outcomes, and both sat in Assur, a thousand kilometers from the goods.

Economic theory and evidence

Larsen, ch. 21: Polanyi and the ancient market

Two cultures mix

- Larsen describes the encounter between Assyrians and Anatolians as **hybridization**, not as one culture imposed on the other.
- Materially, the Assyrians are almost invisible: houses, pottery and objects are all in local style, and only the archives and the seals reveal their presence.
- The Anatolians adopted cuneiform writing and the cylinder seal from the Assyrians, but kept their own political system.
- Mixed marriages became common in the following generation, and children could be given Assyrian or Anatolian names.
- Over time, one can no longer read someone's ethnicity from their name or from the style of their seal.

Religion in daily life

- In the archives, religion is domestic and individual: temples, processions and hymns barely appear, only the religious life of families.
- What remains are practical incantations for moments of crisis: against the demon of the road, against the evil eye and, above all, against the dangers of childbirth.
- Religion also entered trade: the god Assur required votive offerings (**ikribu**), promised funds that stayed active in the business, with the profit going to the temple.
- Failing to pay those offerings was dangerous: it was believed to bring divine punishment and even death.
- Each family had its own gods and venerated its ancestors, buried under the floor of the house.

The problem of chapter 21

- There is no comparable body of texts from any other site or period in the ancient world, the Greek and Roman included.
- But the evidence is one-sided. Only where its light falls are we well informed, and it falls mainly on exchange.
- Nothing survives from Assur, where the money was and the biggest decisions were taken.
- So: can modern concepts such as **market** and **profit** be applied here at all?
- The chapter's epigraph is a German joke: why do something the simple way when it can be done nicely complicated?

Profit as a stated motive

- From a text dissolving a partnership, Ennam-Assur to a former partner:
“In accordance with the policy of the city Assur, any Assyrian will go where it is profitable for him.”
- The phrase can also be read as “where you can make a living”, but the commercial ethos is clear either way.
- Defending a thin return to an investor, the same writer cites the going rate: in Wahshushana and Shalatuwar those doing business realize at least 10 minas of silver for their investors.
- Investor and trader were tied together by an expected **return on capital**.

Markets as places

- *mahirum* carries three senses in the dictionary: marketplace, commercial activity, and tariff, price equivalent or rate.
- Kanesh had a “House of the Market”; the Anatolian side had an overseer of the market, *rabi mahirim*; the market gate is a named place of transactions.
- It is where merchants expected to find each other: “if he does not show up at the market, then have the secretary of the colony search the house of Shat-Ea, the daughter of Suen-dada.”
- Larsen’s caveat: a physical marketplace does not make a self-regulating market. Nobody claims that, and attacking it is a straw man.

Prices moved

- Bargaining is attested directly: “As for the textile you sent me, I have shown it a couple of times. Since people offer a low price for it, your servant Ili-ashranni brings you the textile back, still under your seal.”
- Scores of letters ask an agent to sell at the best possible price, literally “cheap or expensive”.
- Local conditions mattered: “As they are harvesting and commerce has fallen flat, I did not even want to unpack the tin.”
- Veenhof's four price factors: transport cost (about **15 percent** of the value of the shipment), quantity, timing, and information.

Merchants who did the arithmetic

- Inbi-Ishtar, refusing to touch a cheap Anatolian fabric:
“What profit do pirikannu textiles yield that I should trade in them? Assur and Shamash may trample such profit to dung! Thirty donkey-loads are worth only 30 pounds of silver. How many donkey drivers, how much for their harness, and what kind of a journey should they make?”
- A merchant with copper waits out a rival's stock: in ten days the palace supply runs out, then he sells and ships the silver home.
- Costs, volumes and timing, weighed against a margin.

Polanyi's claim

- Karl Polanyi, 1957, “Marketless Trading in the Time of Hammurabi”: the ancient economy was run not by markets but by **reciprocity and redistribution**.
- Applied to Kanesh, three consequences:
 - merchants were traders by status, appointed by authority, living on a commission on turnover, not on price differences;
 - prices were equivalencies fixed by custom, statute or proclamation;
 - the activity was therefore risk-free, and insolvency impossible.
- The claim is not small: it makes the ancient world a different kind of economy, not a poorer version of ours.

Hicks: custom and command

- John Hicks, *A Theory of Economic History* (1969), ch. II, asks what came before the market. His answer: two pure types of non-market economy.
- The **customary economy**: every member performs an allotted task set by tradition. Even the king or chief has prescribed functions. It can last for ages without new decisions.
- The **command economy**: born of an emergency that custom cannot handle, so it is military at first. “The Bantu villages throw up a Chaka, the Mongol nomads a Genghiz Khan.”
- Pure command survives only while the emergency lasts. Afterwards custom creeps back, and the army has to be turned into a civil government.
- Real economies are mixtures: feudalism, where custom has become dominant; classical bureaucracy (Egypt, China), where command stays stronger.

The Revenue Economy

- What the mixtures share: the central economic nexus is **revenue**, the tax, tribute or land rent paid by the cultivator to some authority. Without a market these are not distinguished.
- Revenue is paid in kind. Carrying it to the centre and back is absurd, so local lords take their share on the way. That is feudalism, and the centre withers.
- The only alternative is a bureaucracy with three defences: servants who check other servants, rotation of posts, recruitment of outsiders.
- Concentrated revenue is where specialization first appears: the sculptors of the Pyramid Age were servants of the Pharaohs.
“We have been so accustomed, ever since Adam Smith, to the association of division of labour with market development, that it comes with something of a shock when one realizes that this was not its origin.”

Hicks: where the merchant comes from

- Ch. III. The Revenue Economy has farmers, soldiers, administrators and craftsmen, “but there are no traders, no one who is specialized upon trade.”
- From below: the harvest festival becomes a fair, the peasant with more to sell becomes a middleman, and once he keeps stock at the market-place he has opened a shop.
- From above: a king sends a steward abroad with gifts and instructions about the return gifts. Repeated, the task becomes his function. “He is not an independent merchant, but he is a merchant.”
- The steward takes a commission, trades a little on his own account, and part-time merchanting drifts into whole-time.
- Polanyi's trader by status is Hicks's starting point, not his end point. The question for Kanesh is how far down that road the Assyrian houses had gone.

The test, and Larsen's verdict

- Larsen on the ideas themselves: “entirely speculative and not linked to any concrete evidence.”
- Veenhof, 1972, tested them philologically, working through the key words in hundreds of texts: “to buy”, “price”, “purchases”, “market”.
- His conclusion amounts to a rejection of the whole interpretation. Confronted with the texts, the theory collapsed.
- The dictionary had already given *mahirum* its market senses.
- There was buying and selling, negotiated prices, real profit and real insolvency.

Why the theory survives anyway

- Renfrew and Bahn's standard archaeology textbook still reads the Kanesh merchants as agents, and so as redistribution. Their definition:
"Redistribution implies the operation of some central organization. Goods are sent to this organizing center, or at least are appropriated by it, and are then redistributed."
- Larsen's objection: the merchant houses in Assur organized the purchases and dispatched the caravans. No central organization ran it.
- Aubet keeps "the State" as the great unknown hidden by the missing Assur archive. Larsen: the decisions were taken by a group of men with a personal stake, not by impersonal institutions.
- Even the king sent caravans, with his sons as transporters, depending on agents at Kanesh like everyone else.

The methodological lesson

- Big theories flatten variety. The alluvial plain of southern Iraq and a strategically placed transit city like Assur were not one economy.
- The Anatolian kingdoms were unlike Assur too, and that did not stop the trade.
- Larsen's own method is text-driven, and he grants Mieroop's warning that what was written down is only a fraction of economic life.
- He does not build a model. That step belongs to economists.
- Today's second paper takes it, for the whole Mediterranean.

Of Mice and Merchants

Bakker, Maurer, Pischke and Rauch (2021)

Abstract—We study the causal relationship between geographic connectedness and development using one of the earliest massive trade expansions: the first systematic crossing of open seas in the Mediterranean during the time of the Phoenicians. We construct a geography-based measure of connectedness along the shores of the sea. We relate connectedness to economic activity, which we measure using the presence of archaeological sites. We find an association between better-connected locations and archaeological sites during the Iron Age, at a time when sailors began to cross open water routinely on a large scale. We corroborate these findings at the world level.

The question

- Besides accumulating factors and inventing techniques, exchange and specialization are a classic source of growth.
- If that matters, a place with more potential trading partners in reach should have an advantage.
- The test case is one of the earliest massive trade expansions: the first systematic crossing of **open seas** in the Mediterranean.
- Much early trade was luxury goods for elites, which may have raised their utility without raising productivity. If connections moved settlement, that argues for productivity.
- In the authors' words: "trade seems to have affected the growth of settlements even at an early juncture".

The setting: leaving the coast

- For millennia sailors hugged the coast. Sixty to seventy kilometer crossings were routine in the second millennium BC; longer ones were not.
- From about 900 BC, Phoenician, Greek and Etruscan ships began crossing **300 to 700 km** of open sea.
- Cunliffe's name for the period 800 to 500 BC: "The Three Hundred Years That Changed the World".
- Physical evidence: two Phoenician wrecks 65 km off Ashkelon, about 750 BC, each carrying some 400 standardized wine amphorae. Compare the mixed cargo of the Uluburun wreck a few centuries earlier.
- The Phoenician enclaves (Carthage, Gadir) were nodes in a network, not attempts to dominate local populations.

The mice

- The house mouse occupied a small part of the Levant until the Neolithic revolution.
- By 6000 BC it had reached southern Anatolia, then northeastern Africa and the Aegean.
- No house mice west of Greece by 1000 BC.
- Within a few centuries the animal is on islands and mainland throughout the central and western Mediterranean.
- A biological record of how dense the sea connections had become. Hence the title.

Why the shape of the coast starts to matter

- Sailing close to the shore, coastal shape barely changes how many neighbors are within a given distance. You follow the line either way.
- Once ships cross open water, the shape of the coast and the position of islands decide how many points are reachable.
- Some places come within reach of many neighbors, others of very few.
- That is the variation the paper uses: geography that was nearly irrelevant becomes a **locational advantage** when the technology changes.

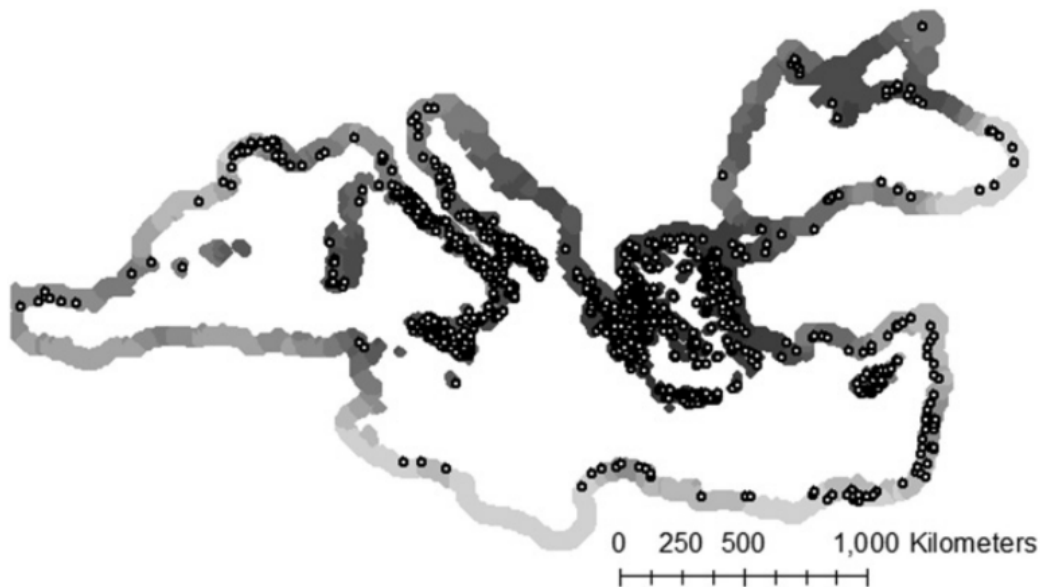
The measure

- The Mediterranean and Black Sea coast cut into a grid of 10 by 10 km cells: 12,013 cells in the sample, 3,352 of them coastal.
- Distance between two coastal cells computed **over water only**.
- Connectedness: how many other coastal cells lie within a sailing distance d , taken as 500 km.

$$c_i^d = \#\{j : \text{sea distance}(i,j) \leq d\}$$

Built from coastline and islands alone. No population, no output, nothing that is itself a product of development. Islands under 20 square km are excluded.

FIGURE 1.—CONNECTEDNESS FOR 500 KM DISTANCE AND PLEIADES NARROW SITES, 750 BC



Reading the map

- Best connected: the island-dotted **Aegean**, and the waist of the Mediterranean at southern Italy, Sicily and modern Tunisia.
- Worst: the straight North African coast, parts of Iberia and southern France, and the Levantine coast.
- Note that the Levant, home of the first cities and the technological leader, scores badly on this measure.
- The 10 km grid gives variation inside countries, not only between them.
- Coarse on purpose: it catches a bay 50 km across, not a natural harbor 5 km across. So it measures reach, not anchorage.

Measuring development without data

- There is no output figure or census for 750 BC. The stand-in is the **count of archaeological sites** in a cell.
- The assumption: more activity means more settlements, and settlements leave traces.
- Source: Pleiades, a gazetteer of ancient places built out from the Barrington Atlas, giving each place a start and end date.
- Narrow definition: entries categorized as urban or settlement. Wide: any man-made structure, excluding natural landmarks.
- For 98.5 percent of cells the count is 0 or 1, so this is close to an indicator of presence.

What can go wrong with the record

- Snodgrass's "positivist fallacy": "making archaeological prominence and historical importance into almost interchangeable terms: in equating what is observable with what is significant".
- Coverage is adequate from about 750 BC and thin before it: 142 sites in 1500 BC against 1,565 in 750 BC and 5,707 in 1 AD.
- North Africa is under-explored, and it also happens to have a straight coast. Results are shown with it dropped.
- Both variables are crude proxies, which pushes against finding any relationship at all.

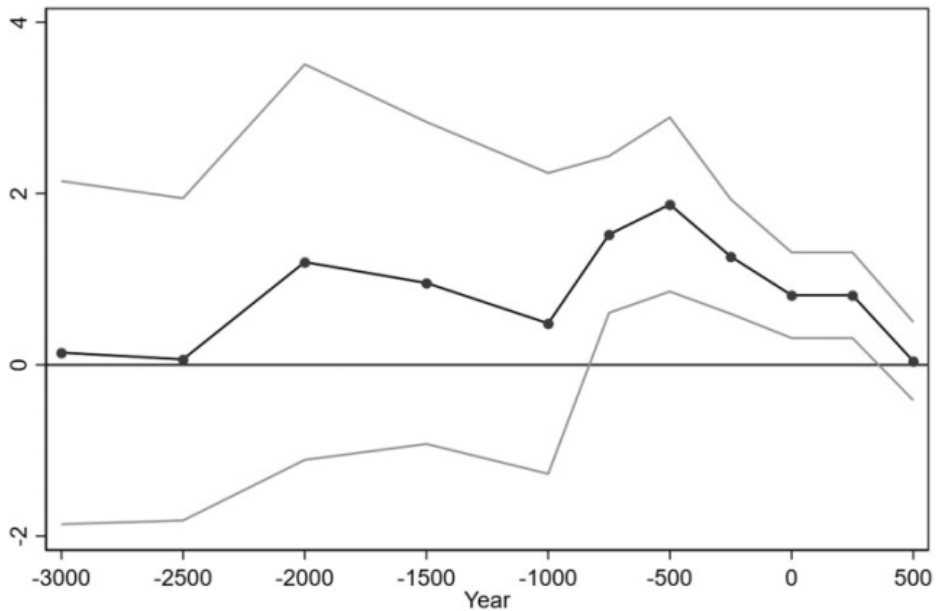
The main result

- In **750 BC**, better-connected cells hold more archaeological sites.
- The response is more than proportional: 10 percent more connections buys more than 10 percent more sites.
- Drop the Aegean and it falls but stays positive and substantial, so the Greek islands are not driving it.
- Drop the coastal cells themselves and keep only the near hinterland: still there. So this is not just the location of harbors.
- Drop connections under 100 km: still there. It is the long-range links, the ones open-sea sailing created, that carry the result.

Is connectedness just good land in disguise?

- Coastal shape has been roughly constant since 3000 BC and owes nothing to development, so it can be treated as given.
- Checked against agricultural yield, ruggedness, distance to rivers, distance to mines, and connectedness over land: no systematic association across subsamples.
- The one exception is wind speed, which does correlate with connectedness.
- Counting reachable **settlements** rather than reachable coast points gives the same picture, and the effect is about twice the size of the market-access estimate for US railroads (Donaldson and Hornbeck).
- What is not observed is the channel: goods, migration, or the spread of ideas.

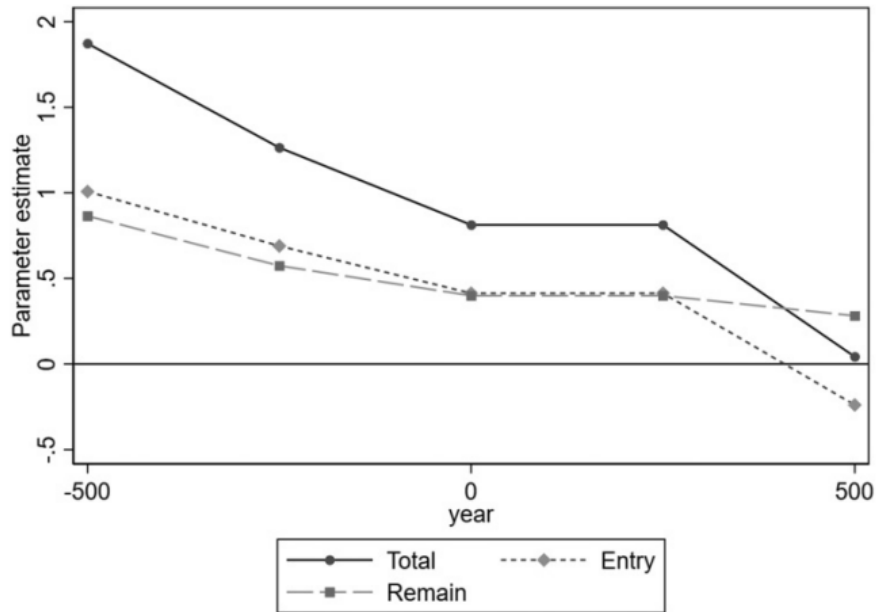
FIGURE 3.—SCALED COEFFICIENTS FOR DIFFERENT YEARS



Reading the time path

- Positive but imprecise through the second millennium BC, when open-sea sailing was rare and the site data are thin.
- It rises at **750 BC**, when open-sea crossings become routine. That is the prediction.
- Then it declines steadily after 500 BC, to nothing by the end of the Roman empire.
- The decline was not expected. The sea stayed central, seafaring kept expanding, and the urban literature finds strong persistence of city locations.

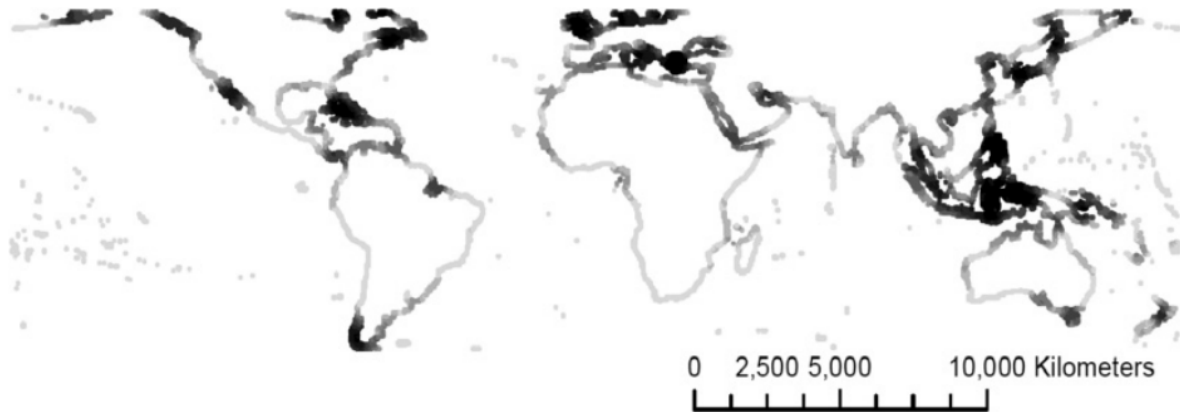
FIGURE 4.—SCALED COEFFICIENTS FOR NARROW PLEIADES SITES: TOTAL, EXISTING, AND NEW ENTRY



Why the relationship fades

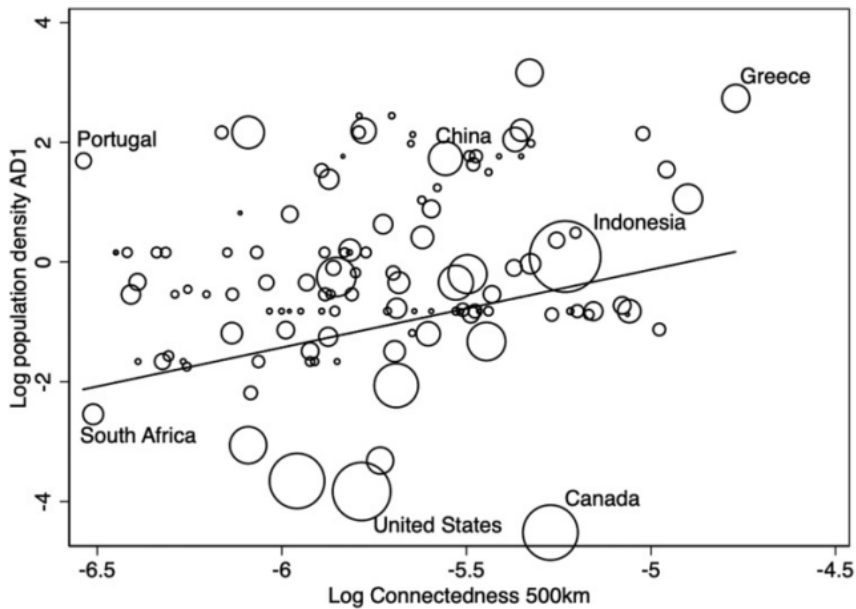
- **Saturation:** 1,565 sites in 750 BC against 5,707 in 1 AD, in only 12,013 cells. The grid fills up and the useful variation goes, since there is no measure of size.
- Splitting the data: sites already present in 750 BC keep the relationship, while sites entering later show a weaker one, eventually negative.
- Reading: the earliest settlements took the best-connected spots; later ones filled in further out, away from existing cities.
- As the new entrants grow as a share of the total, they drag the overall pattern down.

FIGURE 2.—CONNECTEDNESS IN THE WORLD FOR A 500 KM DISTANCE MEASURE



f gray indicate deciles of the connectedness distribution. Darker points show better-connected areas.

FIGURE 5.—CONNECTEDNESS AND POPULATION DENSITY AROUND 1 AD AT THE WORLD SCALE



The world check

- The same measure computed worldwide, on 50 by 50 km cells, against population density in 1 AD for 122 countries.
- Best-connected coasts: Greece again, and Southeast Asia, Chile, Britain. Worst: western Africa and eastern South America.
- The slope is **1.30**, close to what the Mediterranean data imply, though the outcome variable is different.
- Many Mediterranean countries sit in the high-connectedness, high-density corner.
- Portugal, Canada and the United States sit far off the line, which is a reminder of how coarse a country-level measure is.

How to assess it

Geography and institutions together

The two halves of the class

- Larsen shows the machinery exchange needed: treaties with local kings, colony courts, tax rules, partnership contracts, a chain of command running back to Assur.
- Bakker shows that where exchange was cheap is where activity settled, in a period before any of that machinery is visible in the record.
- Neither displaces the other. The Assyrian caravans and the Phoenician ships both worked a given geography, and both needed agreements to work it.
- Larsen's own point about Assur: it was a transit city, strategically placed. Placement first, institutions on top.

An aside closer to home

- Brazilian colonial settlement is a string of ports: Salvador, Recife, Rio de Janeiro, Sao Vicente.
- The interior fills in only where something is worth carrying out, as with the gold of Minas Gerais.
- Same logic as the connectedness measure, with a different sea and a later shipping technology.
- Whether those early advantages still show up in income today is exactly the persistence question the paper raises for the Mediterranean.

This comparison is ours, not the paper's. Larsen does supply one Latin American echo: Tenochtitlan and Tlatelolco as a double city, government beside commerce.

Strengths of the paper

- Market access built out of **pure geography**, so it cannot be a product of the development it is meant to explain.
- A clean timing prediction: the advantage should appear when open-sea sailing appears, and it does.
- Not explained away by the Aegean, by harbors, by soil, ruggedness, rivers or mines.
- The same pattern at two very different scales, the Mediterranean grid and the world.
- It connects an ancient episode to the modern literature on trade and market access.

Limits

- Both variables are proxies. Sites are not output, and connectedness is not observed trade.
- The channel is unidentified: goods, migration, ideas. The authors read “trade” broadly for that reason.
- The pre-750 BC record is weak, and the early sites in the data are largely the ones that survived into the classical period.
- Cross-sectional data cannot rule out that people simply settled where trade was convenient without trade being productive. The authors appeal to Occam’s razor rather than to evidence.
- Larsen’s caution applies here too: the intense light falls on a narrow patch.

Takeaway

- Kanesh shows long-distance trade running on institutions built for it: treaties, colony courts, a ten-year capital contract with transferable shares.
- Those institutions were fragile. Ten deaths in a few years, and a system built on personal credit contracted for a century.
- The Iron Age Mediterranean shows the other force: where the sea made contact cheap is where settlement concentrated, and the advantage held.
- And the ancient world is legible with ordinary economic vocabulary. Larsen documents price, profit and insolvency; Bakker shows the geography of trade leaving a measurable mark.

If there is time

Boehm and Chaney, *Trade and the End of Antiquity*: the same question, a thousand years later, read off ancient coins

TRADE AND THE END OF ANTIQUITY*

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Geneva Graduate Institute

Thomas Chaney[‡]

University of Southern California

January 2026

Abstract

What was the role of trade, and how did economic activity evolve at the End of Antiquity, when political power shifts away from the Mediterranean towards northern Europe and the Middle East? To answer those questions, we assemble a database of hundreds of thousands of ancient coins from the fourth to the tenth century, estimate a dynamic model of trade and money where coins gradually diffuse along trade routes, and recover granular regional trade and real consumption time series. Our estimates suggest that Mediterranean trade was disrupted by the newly formed border between Islam and Christianity; that economic activity shifts away from the Mediterranean starting in the fifth century; that real consumption peaks in the Middle East in the eighth century; and that by the end of the ninth century, Atlantic regions from Islamic Andalusia to Frankish northwestern Europe have become the wealthiest regions of the ancient western world.

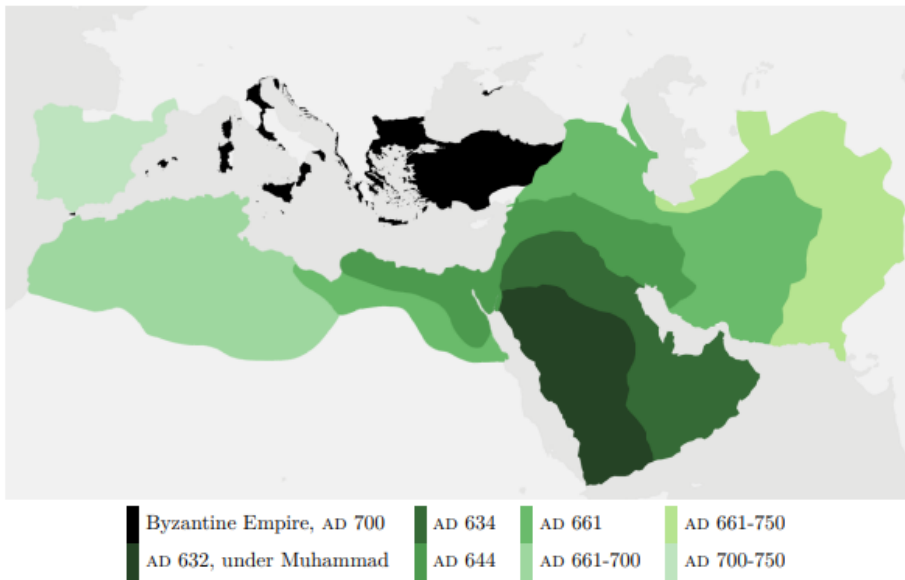
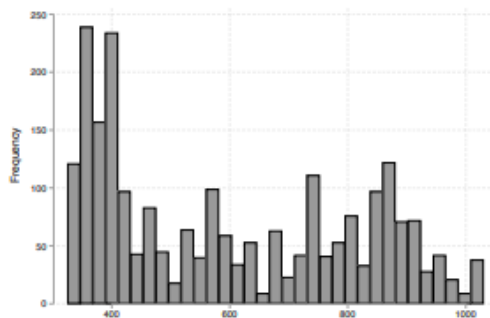


Figure 1: The Arab Conquests, AD 632-750

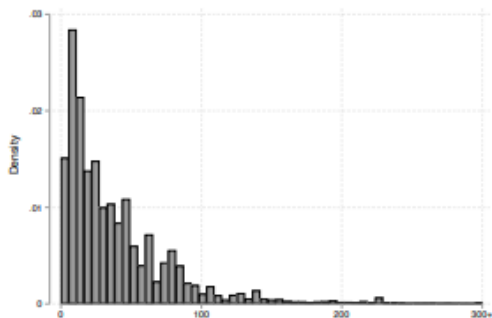
No.	MINT	DATE	DIAM.	WEIGHT	NUMB.
51	الأندلس	114	29.	2.93	4
52	"	115	29.5	2.92	1
53	"	116	26.5	2.92	3

Figure 2: Coin hoard data, an example from al 'Ush (1972)

Notes: The figure shows an excerpt of an original publication from which we assemble hoard data: al 'Ush (1972) gives the content of the Damascus silver hoard in tabular form. From left to right, for the first row: the record number (51), the mint (al-Andalus), the date (year 114 of the Hijri calendar), diameter (29mm), weight (2.93g), and the number of coins with these attributes (4). The issuing dynasty (Umayyad) is given in the table headings and the material and denomination (silver *dirham*) is stated in the text.



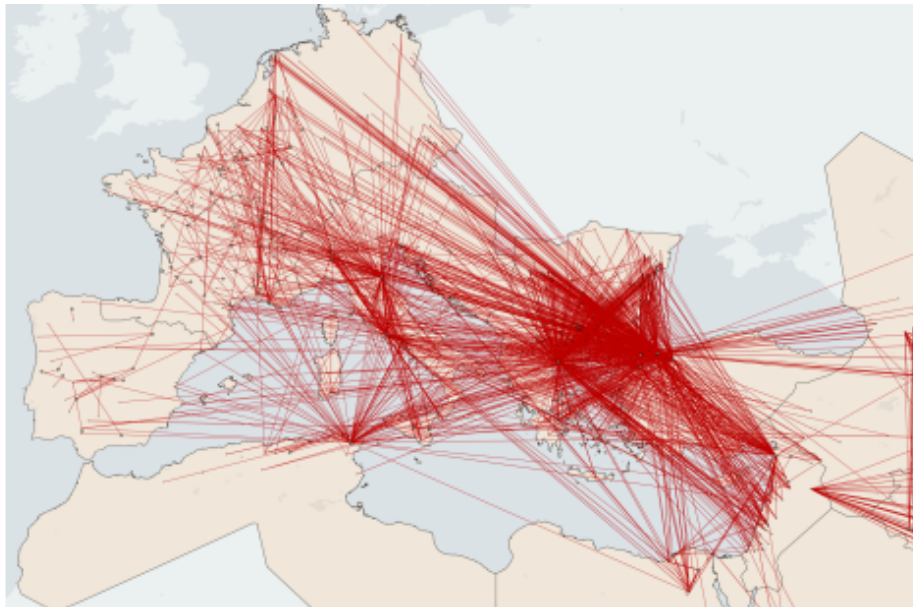
(a) Distribution of hoards by *tpq*, AD 325-950



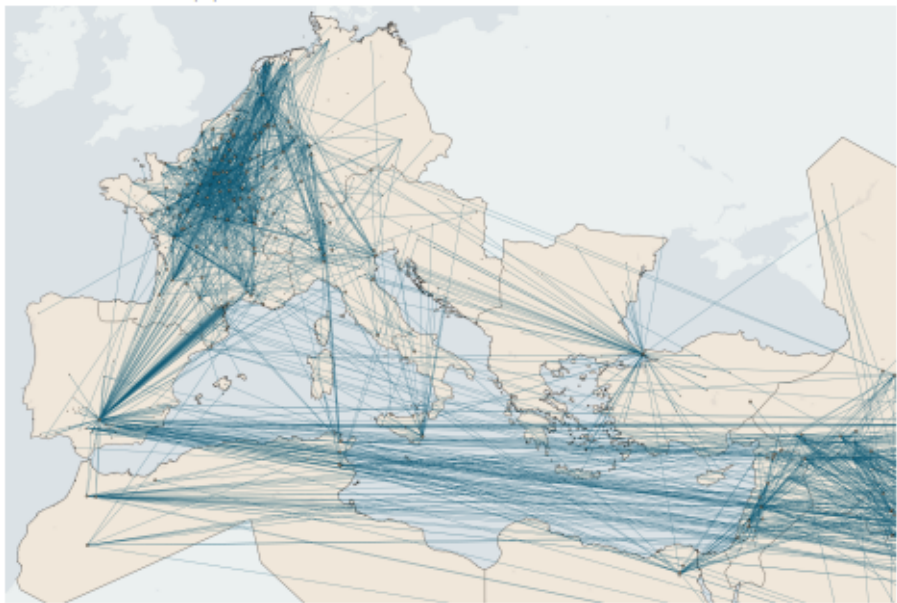
(b) Coin age at time of deposit (*tpq*), in years

Figure 3: Number of hoards over time and ages of coins within hoards

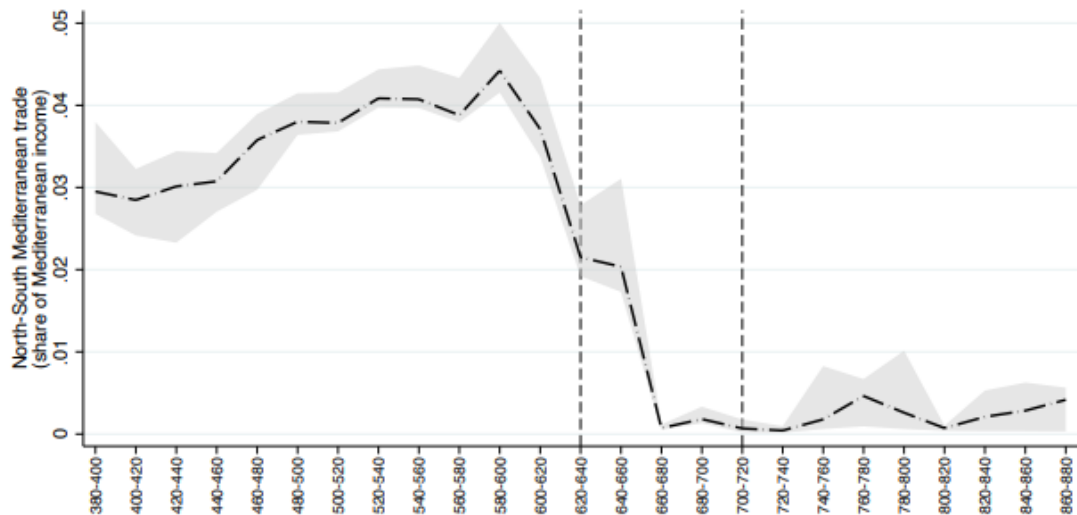
Notes: Panel (a) shows the number of hoards per 20-year period. Panel (b) displays the (annual) density of coins of different ages. Coin age is defined as the difference between the midpoint of the coin's minting interval and the *tpq* of the hoard it is found in.



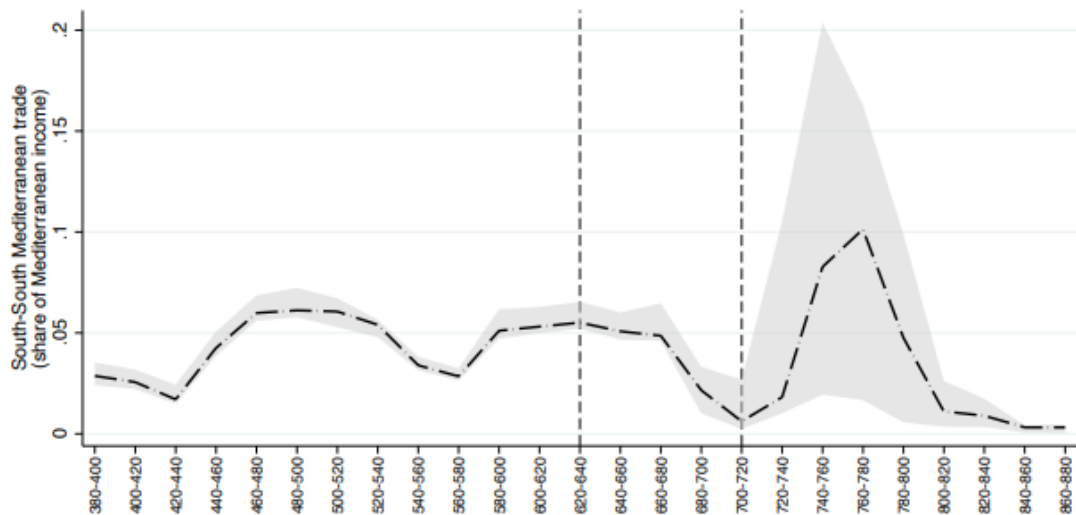
(a) Before the Arab conquests: AD 450-630



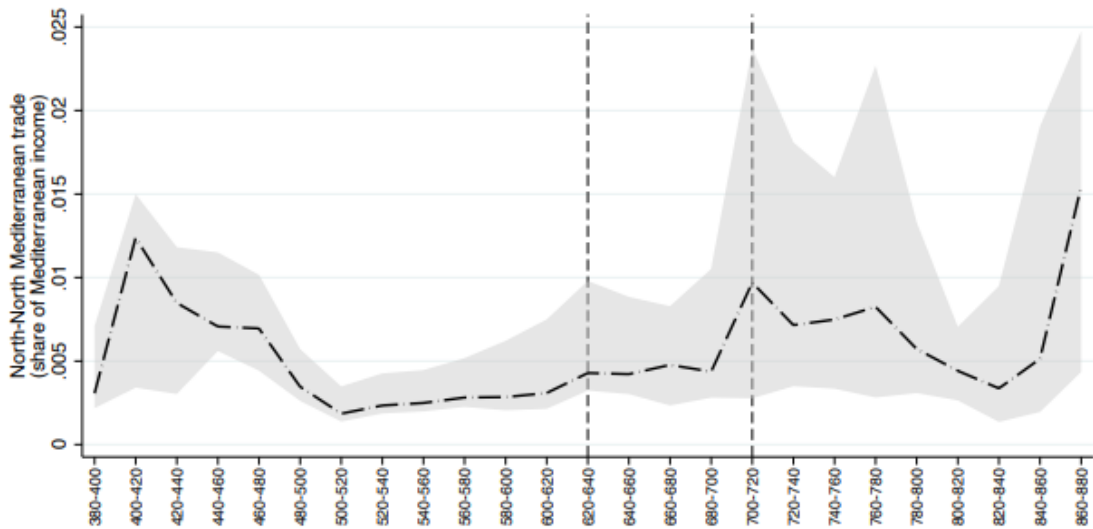
(b) After the Arab conquests: AD 713-900



(a) North-south trade



(b) South-south trade



(c) North-north trade



(a) Relative real consumption changes: pre- to post-conquests



(b) Urban Population Growth in European Regions, AD 700-900

Figure 11: Real consumption and urbanization